SAINT MICHAEL COLLEGE OF CARAGA

IMPACT OF OPLAN LIGTAS NA PAMAYANAN ON FIRE SAFETY BEHAVIOR OF HOUSEHOLDS IN BUENASCAR, AGUSAN DEL NORTE

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the Study

Fire incidents continue to be a serious threat to people's safety in many Filipino homes. Among the causes are poor electrical wiring, a lack of understanding about fire safety, and the use of lightweight or easily flammable materials in home construction. Unprepared households are especially at risk because even small fires can cause severe property damage, fatalities, and lasting social and economic problems for families and local governments Paulino & Matthew Paulino, (2023).

The Bureau of Fire Protection (BFP) created and carried out Oplan Ligtas na Pamayanan (OLP), a nationwide community-based fire prevention program, to raise awareness of fire safety issues and promote safer barangay practices in response to these threats. Through community fire safety education, house-to-house inspections, and residents' active involvement in various risk-reduction initiatives, the program seeks to lower the risk of fire. (BFP, 2022). The program was basically focused on urging households to use proactive fire safety precautions.

The program aimed to encourage households to adopt proactive fire safety measures, including proper handling and safe storage of flammable materials, correct use of electrical appliances, and preparedness for emergencies Capiña, (2022);Macarubbo, (2022). OLP has become a vital strategy for lowering the incidence and consequences of fires in various Philippine communities over time Lagumbay et al., (2023). Currently, no empirical evidence exists on how OLP affects household fire safety practices in smaller communities, particularly in Buenavista, Nasipit, and Carmen, Agusan del Norte. This lack of local data matters because community programs may function differently depending on participation, income, and local risk factors.

By investigating the effects of Oplan Ligtas na Pamayanan on fire safety practices in homes in Buenavista, Nasipit, and Carmen, Agusan del Norte, this study seeks to close the gap. The study will give the Bureau of Fire Protection, local government entities, and community leaders useful information by evaluating the program's Efficacy in homes. The researchers expect that the results will help guide policy decisions and improve fire management systems, ensuring that fire safety practices in the community are strong and lasting.

The Fire and Disaster Management Agency (FDMA) in Japan is responsible for fire safety and emphasizes neighborhood-based initiatives. These include information campaigns, public drills, volunteer organizations for disaster prevention, and routine inspections FDMA, (2024) International organizations have praised Japan's disaster management systems for their practicality and organization GFDRR, (2013). Similarly, South Korea's National Fire Agency (NFA) manages various fire safety initiatives, including public awareness campaigns, inspections, and training. Meanwhile, China's fire safety system has developed rapidly with the country's industrialization and urbanization, and recent studies indicate that fire regulations and management practices have improved Zhang et al., (2019) The National Fire Fighting Corps (NFFC) implements fire safety measures, including routine inspections and public education campaigns Zhang, (2021);Liu et al., (2023). These examples show that community-based approaches are an important part of successful fire prevention worldwide.

The Philippines' tropical climate, especially the high temperatures during the dry season, increases the risk of fire. Dense residential areas and rapid urbanization further increase the country's vulnerability to fire-related disasters Manalo et al., (2022). Fires often destroy homes and property, cause fatalities, create economic problems, and slow down community development. Ann et al., (2022) Because of this, there is a pressing need to improve fire safety and preparedness efforts across the nation.

Many households still do not follow basic fire safety measures, even though fire safety programs exist. While significant buildings usually have sprinklers and other safety systems, some homes do not have basic fire equipment, like fire extinguishers Albano, (2024). Securing a home's electrical wiring reduces the risk of fire. However, many homeowners attempt to save money by doing their own electrical work, which often leads to unsafe or poorly connected wiring.

Surveys conducted in Bislig City supported the findings of Martin's Electrical and Lighting Company, (2025), which revealed that many households used incorrect fuse ratings, neglected to have their wiring inspected regularly, did not adhere to clear fire safety procedures, and had little knowledge of how to prevent electrical fires. Concrete homes were more likely to have fire extinguishers, proper permits, and better safety practices Nocete & Enteria, (2020).

According to specific research, being aware of fire safety precautions does not guarantee that people follow them. According to one review, although many homes have fire safety equipment, people often neglect it or are unsure how to use it in an emergency Senthilkumaran et al., (2019) Household negligence and a lack of accountability can reduce the effectiveness of safety measures. Age-related differences in perceptions of safety are also significant. For example, younger people are more careful about fire safety, while older people often take risks, such as leaving appliances on overnight Aliesa Mohd Zul et al., (2024).

This study aims to understand fire safety behavior better and to highlight how the Bureau of Fire Protection's awareness programs help increase fire safety knowledge and preparedness in the community. The study examines explicitly how Oplan Ligtas na Pamayanan (OLP) affects home fire safety procedures, emphasizing participation in risk reduction, awareness, and preparedness. The study will ascertain how community-based initiatives, such as inspections, lectures, drills, and risk assessments, affect household behavior in reducing fire hazards by assessing the program's Efficacy.

Several groups could benefit from the study's conclusions. The findings can help households increase their awareness and practical understanding of fire safety. Barangay officials can use the information to improve their community safety initiatives and information campaigns. The Bureau of Fire Protection will use evidence-based insights to improve current programs and policies. Additionally, this study will add to the body of knowledge regarding community-based fire prevention initiatives and their importance for disaster preparedness in local communities.

1.2 Theoretical Framework

PROTECTION MOTIVATION THEORY

Rogers first proposed the Protection Motivation Theory (PMT) in 1975, and he later refined it in 1983 to describe how persuasive communication influences behavior Rogers, (1975).It highlights the reasoning behind deciding whether to follow or disregard a suggested course of action. The degree to which BueNasCar households understand the repercussions of fire incidents and the probability that they pose is referred to in this study as threat appraisal. Coping appraisal, on the other hand, looks at people's confidence in using these measures, their belief that the measures can prevent or reduce fire damage, and any obstacles they might face in doing so.

The Protection Motivation Theory (PMT) states that when households believe the program's safety measures are effective, feel confident in implementing them even in the face of challenges, and view fire as a serious and likely threat, they are more likely to adhere to fire safety practices. PMT is a helpful way to understand and predict fire safety behavior in households targeted by Oplan Ligtas na Pamayanan. Studies in the Philippines support this framework, showing that self-efficacy and response efficacy are strong predictors of disaster preparedness Kurata et al., (2023).

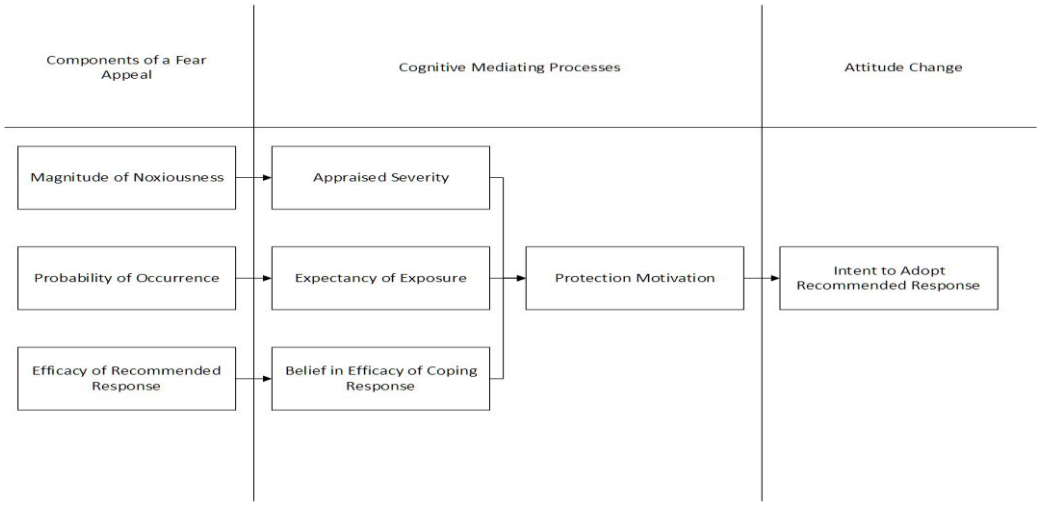


Figure 1: Protection Motivation Theory

The Protection Motivation Theory framework, which shows how households evaluate the effectiveness of the program's fire safety measures, is shown in Figure 1. These measures include taking part in community fire drills, installing smoke alarms, and storing flammable materials safely. During the 'Cognitive Mediating Processes' stage, people think about the different parts of the fear appeal.

The term' appraised severity' refers to how people judge the possible damage a fire could cause in their neighborhood. Expectancy of Exposure measures how at risk people feel they are from a fire. Belief in the Efficacy of Coping Response shows how confident they are that the actions recommended by Oplan Ligtas na Pamayanan can prevent or reduce fire damage. Their confidence in Oplan Ligtas na Pamayanan's suggested actions to prevent or reduce fire damage reflected in their belief in the effectiveness of the coping response. The model's 'Attitude Change' element also supports the program's goal of encouraging families to follow recommended fire safety practices. These include joining community fire drills, using electrical appliances safely, and participating in local fire departments.

The PMT is one of the most influential theories explaining individual risk prevention and the adoption of protective actions. First, the theory was used to explain how fear affects individuals and individual-related health adoption behaviors. It highlights that protective motivation arises from people's desire to avoid potential adverse outcomes following a perceived threat (Rogers, 1975). It has also been used over time to explain self-defense measures in a variety of emergency scenarios Grothmann & Reusswig, (2006). Protective motivation, or the desire to take preventive measures against potential threats, is assumed by PMT to be a process that combines threat assessment and response appraisal. The formation of a protective motivation, which in turn affects behavioral change, is determined by these results.

According to the theory, being willing to alter one's behavior is the best defense. Researchers define threat appraisal as people's awareness of risk factors, specifically their perceived vulnerability to and the severity of a disaster. These evaluations combine to form Protection Motivation, the mental motivation to practice fire safety.

According to Rogers, (1975) and Floyd et al., (2000),these assessments in PMT are multiplicative: even if residents have strong beliefs, their motivation will decrease dramatically if any one of them is weak.

1.3 Review of Related Literature

In the Philippines, fire incidents continue to pose a significant threat, causing fatalities, property damage, and livelihood disruptions. The BFP has responded by implementing national fire awareness campaigns, including school-based fire drills designed to minimize injuries during emergencies. Opiana, (2017)

Given its direct impact on property and human lives, ensuring fire safety is a top priority Gautami et al., (2020). In response to the rising number of fire incidents across the country, the Bureau of Fire Protection (BFP) launched a comprehensive awareness campaign in 2019 to

commemorate Fire Prevention Month, as required by Presidential Proclamation No. 115-A. The program aimed to establish fire-safe communities that would enhance each Filipino's overall quality of life.BFP, (2022)

Establishing efficient fire safety procedures is crucial to protecting people and property in a nation highly susceptible to hazards. Beyond being legally mandated, following the Philippine Fire Code is also morally important to save lives and property. Lagata et al., (2022). Effective fire prevention requires regular risk assessments, reliable detection and suppression systems, and proper maintenance of fire safety equipment. Community education and public awareness are equally essential in preventing fire incidents Fu et al., (2024). Fire risk assessments involve identifying vulnerabilities, recognizing potential hazards, and implementing precautions to reduce risks. By being proactive, both individuals and organizations can prevent fires before they start.

The BFP divides OLP into four categories: "Lingap" (Protective Care), "Handang Pamayanan" (Ready Community), "Bayanihan" (Cooperative Undertaking), and "Kaagapay" (Companion) BFP, (2022). The Handang Pamayanan program helps communities develop hazard maps and fire safety plans. In contrast, the Bayanihan program trains volunteer brigades to join the Community Fire Auxiliary Group (CFAG) and supports neighborhood projects. The Kaagapay program fosters skill development through barangay visits and home fire safety exercises. The Lingap program honors barangays that conduct thorough house-to-house fire safety surveys, especially in densely populated, high-risk areas BFP, (2022). Together, these programs encourage planning, readiness, and community involvement, effectively reducing the number of fire incidents Rosales, (2019).

While BFP teams are en route, the CFAG, a formal volunteer organization composed of locals trained through the Bumbero sa Barangay program, responds to local fire incidents. This group plays a crucial role in enhancing neighborhood safety and strengthening barangay

firefighting capabilities. Managed by local officials, the program ensures sustainability and efficient coordination across all aspects of fire response. According to Muico et al. (2024), communities demonstrate a strong commitment to fire safety, with residents actively learning and adhering to safety precautions, highlighting the Efficacy of fire safety education initiatives.

Global best practices emphasize awareness and preparedness as key to reducing fire risks. The University of Texas at Austin promotes fire safety by teaching safe cooking techniques, proper fuel handling, correct use of alarms, and escape route planning. Since residents may have as little as two minutes to evacuate during a fire, the United Ways of New Jersey urges families to practice their fire escape plans regularly. According to Addai et al., (2016)Unsafe electrical wiring by unskilled artisans in Ghana often causes short circuits and fire incidents. Open flames from lanterns, stoves, and cigarettes also remain primary sources of ignition. These incidents underscore the need for improved infrastructure, stricter safety regulations, and more effective global awareness campaigns.

Age

Age is an important factor that affects how people see and respond to fire safety programs. Research shows that older adults are generally at a higher risk of fire than younger people because of their life experiences and greater vulnerability Karemaker et al., (2021). Research in Spain shows that older people are more likely to die in fires, partly because they have limited mobility and cannot act quickly enough to protect themselves during emergencies Fernández-Vigil & Echeverría Trueba, (2019). Younger people, on the other hand, tend to be more confident, but they may underestimate risks and be less careful with safety measures Rahman et al., (2021). These differences show how age affects people's views of risk and their safety behaviors.

Age affects both how people think about risk and their physical vulnerability. An extensive study in Sweden found that older adults have a higher risk of residential fire injury compared to

younger adults (Jonsson et al., 2016). The National Institute of Standards and Technology also reported that older adults' frailty contributes to home fire deaths, as they often take longer to evacuate because of reduced mobility and slower reactions.

Disaster education today focuses on ways to help children become more resilient and share information to lower the risk of disasters at home, because children are the most vulnerable group in society Ronan et al., (2015). Some studies' findings indicate that people never forget what they learn early in life, emphasizing the importance of teaching children risk-reduction and disaster-prevention techniques Cvetkovic, (2019). Similarly, Aliesa Mohd Zul et al., (2024) Found that young people know more about fire safety, while older people tend to keep old habits, like leaving appliances on at night, which can lead to fires.

Gender

Regarding Gender, Quiroz et al., (2021) Found that while men are more likely to take action during emergencies, women in informal settlements are frequently quicker to identify fire hazards and alter household practices to stay safe.

According to Sharan & Gaillard, (2025) Institutional and cultural barriers make it particularly difficult for gender-diverse groups to participate in disaster preparedness in both India and the Philippines. Mangahas et al., (2018) found that, despite having limited resources, economically disadvantaged Filipino women take an active role in disaster risk management in their communities by promoting household preparedness and organizing neighborhood safety measures. These studies show that in the Philippines, men usually handle technical or response-focused roles, while women are generally more focused on preventive measures.

While women concentrated on evacuating family members, men's first instinct was to locate the fire's source. According to a separate study by Hahm et al., (2016) people with experience in fire risk situations—either as firefighters or victims—have a relatively simple

understanding of potential fire hazards. The inquiry conducted by Fernandez et al. (2018) found that the perceived fire threat was strongly correlated with age, gender, and household ownership. According to the study, household ownership and age both increase the seriousness with which one comprehends the threat.

Attitude

Their attitudes toward fire safety consistently influence people's intentions and actual preparedness behaviors. Research has demonstrated that people are more likely to take home safety actions, such as installing alarms or creating escape routes, when they have positive attitudes influenced by social influences, past knowledge, and the perceived value of safety precautions. Along with risk perception and self-efficacy, research on fire safety behavior also finds that attitude is a critical component. V. Cvetkovic, (2021).

According to the Bureau of Fire Protection (BFP), unsafe LPG use, unattended cooking, and defective wiring are common causes of fire. The correct information and safe procedures can reduce these risks. Despite laws and programs, many people, particularly business owners, are poorly informed about fire safety measures, thereby increasing the risk of fire-related losses Nudzor, (2023).

Furthermore, residents often take action to put out fires before authorities arrive, showing the importance of community involvement in fire management Ardani & Permana, (2019). Working together helps connect and empower everyone involved in fire prevention, allowing fires to be managed more effectively and efficiently Menya & K'akumu, (2016). According to research, attitudes, confidence, and risk awareness are among the ways fire safety knowledge influences household behavior and functioning. Specifically, Ferrer, (2021) discovered that some towns had high ratings for their fire prevention programs; however, research rarely examines whether officials' expertise or other local factors are responsible for these outcomes. This disparity is

significant because new research shows that community involvement and local leaders' technical and procedural expertise are both necessary for successful disaster preparedness, the United Nations Office for Disaster Risk Reduction.

Additionally, Gandawali & Abdul, (2017) showed a strong correlation between respondents' knowledge of fire safety and the execution of the Oplan Ligtas na Pamayanan program. The Bureau of Fire Protection created an enhancement program to support and reinforce current fire prevention efforts. The study also demonstrated that greater awareness was associated with higher levels of program implementation, underscoring the program's beneficial impact on knowledge of fire prevention.

According to Kurata et al., (2023)Prior knowledge, perceived severity, vulnerability, and personal experiences all influence people's attitudes. People with more positive attitudes are more likely to value prevention programs and take part in safety measures.

Similarly, Muico et al., (2024) evaluated home fire prevention awareness and practices in Barangay San Roque, Municipality of Maco. The findings demonstrated a high level of comprehension and adherence to fire safety precautions, including installing safety equipment and preparing the home for fire emergencies. The study indicates strong engagement in fire safety procedures and a favorable attitude toward fire prevention, despite not being conducted over an extended period. This favorable attitude likely resulted from the BFP's education and awareness initiatives.

In the Philippines, Kurata et al., (2023) found that people's views on the effectiveness of safety programs are closely linked to their positive attitudes toward fire prevention, especially when they understand the risks of fire. Their results provide evidence that awareness-raising, community drills, and active home participation programs, such as OLP, can lead to behavioral and attitudinal changes. This aligns with the Protection Motivation Theory, which holds that when

people are aware of the seriousness of a threat and the effectiveness of preventive measures, they are more inclined to take protective actions. Rogers, (1975).

Self-efficacy

People with special needs and vulnerabilities, such as children, the elderly, people with disabilities or medical conditions, and those who are underserved or have limited resources, have been demonstrated to be better prepared for disasters when they have self-efficacy Hamann et al., (2016); Marceron & Rohrbeck, (2019); Rivera, (2020). Additionally, Adams et al., (2017) found that self-efficacy can influence the relationship between preparedness and perceptions of health. According to Marceron & Rohrbeck, (2019) Self-efficacy strengthens people's perceptions of threats, thereby increasing the likelihood that they will adopt disaster preparedness measures. Research also showed that there is no meaningful connection between threat perception and preparedness for disasters when self-efficacy is low. In other words, if people lack self-efficacy, they are less prepared even when they see a high level of threat. Self-efficacy is often lower in families with members who are especially vulnerable.

Guevara, (2025) defines fire safety as the steps, rules, and methods used to prevent fires, lower the chances of fire-related incidents, and protect people and property if a fire occurs. To stop fires and lessen their effects, a mix of awareness, readiness, and doable safety precautions is needed. Providing fire extinguishing equipment, such as sprinklers, smoke detectors, fire alarms, and fire extinguishers, is part of being prepared to respond to fire disasters. These efforts decrease potential risks, protecting people more effectively. It also means making sure residents know the fire department's phone numbers and have the skills to use the equipment and alarms. Checking that fire exits are easy to reach during a fire is a common way to measure a building's safety, Adeleye et al., (2020).

Only when people have reasonable expectations of their capacity to carry out the act will they develop intentions to prepare for disasters Adams et al., (2019). The majority of research has shown that high self-efficacy can drive specific protective measures in emergencies, like emergency evacuation Tang & Feng, (2018); Yu et al., (2020), as well as disaster preparedness intentions or actual behaviors Samaddar et al., (2014).

Risk Perception

Age and education, as well as fire risk perception and knowledge, were found to be significantly correlated in a study examining the effects of several sociodemographic factors, including age, education level, gender, and prior fire experience Chan et al., (2018). However, Kinateder et al., (2015) state that risk perception is an individual's subjective assessment of the likelihood of a fire and the potential severity of its effects, including feelings of vulnerability and danger, as well as emotional responses. People are more likely to take preventive steps, such as checking electrical systems or keeping fire extinguishers ready, if they think a fire is severe and likely to happen.

In a similar vein, Kurata et al., (2023) found that perceived severity and vulnerability directly influence Filipinos' fire safety practices, two key components of risk perception. They demonstrate that people are more likely to take preventative measures when they are aware of the risks posed by fire.

Expanding on this notion, theories such as the Protection Motivation Theory (PMT) describe how people are motivated to take protective measures based on their perceptions of risk. Perceived susceptibility—the likelihood of being affected—and perceived severity—the gravity of the consequences—are the two primary components of risk perception, according to these models. These components affect a person's intention to implement safety measures when paired with other elements, such as self-efficacy and belief in the Efficacy of the response R. A. Ferrer &

Klein, (2015). People are more likely to take precautions when they perceive a high risk, but they are also more likely to act carelessly or in an unsafe manner when they perceive a low risk. Research on fire and disaster safety found that residents' views of their risk of fire injury are closely linked to their past fire experiences and their trust in the city's fire response services Nilson & Mankell, (2022). This suggests that residents should install fire extinguishers, inspect electrical systems, and take other preventive measures to reduce fire risks, and that households that understand the gravity and likelihood of fire hazards are more likely to participate in community fire drills.

Knowledge by itself does not always translate into safe behavior, according to empirical data. When people feel that they are in danger, they are more likely to take action. For instance, Mccaffrey, (2015) found that when families felt more personally at risk, they were more likely to keep smoke alarms operational and make evacuation plans. Barangays that actively implemented OLP demonstrated higher levels of awareness and readiness. Still, disparities remained, impacted by residents' perceptions of their susceptibility to fire risks, Nanta & Pisueña, (2025). Even with information, some households did not buy fire extinguishers or unplug appliances before leaving, because they believed a fire was unlikely. These results imply that information alone is insufficient. To promote the adoption of workable safety measures, effective risk communication must also account for people's attitudes.

House-to-House Fire Safety Survey

The primary method for carrying out zonal interventions is the house-to-house fire safety survey, which the OLP Lingap Program requires. Under the direction of designated Bumbero sa Barangay staff, survey teams assess each home's compliance with fire safety regulations, identify structural and electrical hazards, and offer suggestions for prompt remedial measures.

Coordination with barangay officials, recording of hazards found, and submission of combined reports to the BFP District Office for oversight and assessment are all part of each survey.

The Lingap Program's house-to-house fire safety surveys are another way that the OLP emphasizes the value of routine inspections. Using this method, inspectors directly inspect residential areas to find hazards, track compliance, and promote prompt risk reduction. BFP, (2022). The program promotes accountability and guarantees that fire safety procedures are tracked and continuously enhanced at the local level by integrating inspections into house calls. Home Fire Safety Visits (HFSVs) or Home Fire Safety Checks (HFSCs) are proactive community-based strategies that fire services use to assess domestic fire risks, increase awareness, and promote preventive behavior Waring et al., (2024).

During these visits, trained firefighters or community volunteers usually inspect homes for hazards, ensure smoke alarms are in place, and instruct residents on safe electrical practices and evacuation planning Hampton et al., (2025). Reducing fire incidents and the injuries and deaths they cause is the goal of several global initiatives. This involves educating and training parents, children, and high-risk groups such as the elderly. Jonsson et al., (2016) & Wuschke et al., (2013). These personal visits provide customized guidance and equipment fittings based on the property layout and obvious hazards. The realization that socially disadvantaged people frequently lack the funds to purchase fire safety equipment is one of the factors driving this. In a similar vein, elderly or disabled people might have trouble setting up equipment or require more specialized guidance on potential hazards and escape routes Jonsson et al., (2016).

O. Ferrer, (2021) Noted that fire safety inspections and assessments were carried out correctly in Aparri. However, some establishments still broke the Fire Code, creating hazards. As a result, authorities advised BFP Aparri to monitor non-compliant buildings more closely.

Zonal Interventions

The zonal interventions concept under the Bureau of Fire Protection's (BFP) Oplan Ligtas na Pamayanan (OLP) framework represents a systematic, community-based approach to fire prevention and preparedness. According to the 2023 Revised OLP Guidebook, zonal intervention is the process of assigning barangays or communities to one of four fire vulnerability zones—red, orange, yellow, or blue—based on the risk levels determined by the BFP, (2022) and the Community Fire Protection Plan (CFPP).

This zonal classification decides how often and how intensely interventions are carried out, such as hazard removal, evacuation drills, fire safety education, and house-to-house fire safety surveys. OLP enables the BFP to effectively allocate resources and ensure that communities at high risk receive more stringent fire prevention measures by classifying areas according to their vulnerability.

The Guidebook states that the Red Zone indicates the most vulnerable areas, such as those with poor housing conditions or a high frequency of fire incidents. Twice a year, fire officers and community volunteers must perform house-to-house fire safety surveys in these areas, along with hands-on hazard removal and informative seminars. While the Yellow Zone, which has low fire vulnerability, needs interventions every two years, the Orange Zone, which has been classified as moderately vulnerable, requires surveys and lectures once a year. The Blue Zone, which is considered reasonably safe, emphasizes maintenance, evacuation planning, and ongoing monitoring. BFP, (2022). The zonal intervention approach is a strategic plan that spreads fire safety initiatives relatively evenly while focusing on risk. This is shown through an organized, color-coded classification system for all barangays.

Theoretically, risk-based targeting and community-based disaster preparedness frameworks, which support directing resources to areas with concentrated risks, also lend support

to the implementation of zonal interventions Waring et al., (2024) found that home fire safety visits in targeted areas significantly reduced household fire risks and gave good cost-benefit results.

Fire Safety Lectures and Seminars

Systematic programs like those described in the Bureau of Fire Protection's (BFP) Oplan Ligtas na Pamayanan (OLP) Guidebook help to strengthen fire safety initiatives at the home and community levels. Fire safety education is an important part of prevention efforts. Households can learn about fire risks and safety measures through programs like community fire safety surveys, seminars, and lectures, as described in the OLP Guidebook BFP, (2022). As previously mentioned, education is an essential tool for raising awareness and guiding families toward safer practices, reinforcing the idea that communities with greater knowledge are more fire-hazard-resistant. The OLP emphasizes how campaigns help foster a culture of long-term fire safety. The authorities urge barangays to carry out zonal interventions, disseminate educational materials, and use social media platforms to increase advocacy through public fire education campaigns BFP, (2022). These campaigns help the BFP, and the community maintain awareness, promote positive attitudes, and consistently apply preventive measures together.

Households that received structured fire safety education demonstrated significant improvements in knowledge and preventive behaviors Hassan Tawfik et al., (2019). This suggests that awareness alone can bring about change when combined with favorable attitudes toward safety procedures. Furthermore, Nilson & Bonander, (2020) emphasize that knowledge heightens perceptions of risk. Families aware of fire risks were more likely to recognize their own vulnerability, which encouraged them to take precautions, such as installing alarms and rehearsing evacuation procedures. Nanta & Pisueña, (2025) found that barangays that regularly attended fire safety seminars followed local fire codes more closely and showed a stronger sense of community

responsibility for preventing fires. This is consistent with the BFP's belief that education empowers people and leads to behavioral change. According to Espartero, (2025), OLP lectures and seminars were considered by the people of Antique Province to be one of the most successful elements of the program because they promoted collaboration with nearby fire stations and improved knowledge of appropriate fire prevention behavior.

Research, however, indicates that safe practices are not always the result of knowledge. Gilbert, (2021) found that because residents try to control fires on their own rather than following evacuation protocols, many home fires go unreported. Due to perception and mobility issues, older adults frequently adopt preventive measures, such as smoke alarms, but often lack comprehensive knowledge of risky behaviors Karemaker et al., (2021).

Drills and Simulation

Building community skills and readiness through training and drills is another key part of OLP. The Guidebook BFP, (2022). describes organized programs like Technical Skills for Responders (TSR) and Basic Skills for Responders (BSR), which teach volunteers and Community Fire Auxiliary Group (CFAG) members practical firefighting and emergency response skills. BFP, (2022) The BFP, (2022) emphasizes that simulation exercises and fire drills are important for testing preparedness, especially in high-risk residential areas. These exercises not only improve individual skills but also boost group confidence and coordination during emergencies.

Nickerson, (2021) found that fire safety training gives communities the skills to respond to fires safely and confidently. Additionally, it imparts preventive knowledge, empowering communities to eliminate risks before they materialize. Both fire safety protocols and procedures for evacuating and remaining safe in the event of a fire should be covered in training.

According to the Guidebook, the purpose of fire drills and simulation exercises is to assess the community's preparedness to handle fire situations. Residents' involvement in evacuation and

suppression protocols, the promptness of responses, and the efficiency of coordination among community fire auxiliaries are all evaluated. The OLP Guidebook recommends conducting these drills at least once a year, especially in communities identified as Red Zones or high-risk areas by the OLP's risk mapping. Residents can practice reporting fires, carrying out evacuation plans, and collaborating with firefighters and barangay responders by using a simulation BFP, (2022). Nickerson, (2021) found that fire safety training gives communities the skills to respond to fires safely and confidently. It also provides preventive knowledge, helping communities reduce risks before they happen. Training should include both fire safety rules and procedures for evacuating and staying safe during a fire.

Involving local leaders, schools, businesses, and vulnerable groups like the elderly and children is encouraged by the Fire Safety Educator and the Community Fire Auxiliary Group (CFAG). This teamwork approach makes sure the community knows its responsibilities during emergencies.

The OLP Guidebook also helps teachers adjust drills to match the specific risks in each sitio or purok, such as crowded homes, narrow pathways, or limited access to water. This localized approach enhances the drills' Efficacy and realism.

V. Cvetkovic, (2020) asserts that citizens should be trained in the use of fire extinguishers in order to increase safety and get them ready for fire. Experts advise keeping flammable materials out of attics, basements, and hallways in homes and buildings as a preventive measure. All electrical appliances should be turned off before leaving the house, especially when the occupants will be away for an extended period. Additionally, it is crucial to regularly maintain evacuation routes.

Fire Preparedness and Planning

Reducing losses while preserving safe built environments and promoting sustainable disaster management techniques is the aim of FDP Kihila & Kihila, (2017). Records of international fire incidents, such as the London Grenfell Tower fire in 2017 and the Shanghai fire in 2010, demonstrate that preparedness and causes differ across countries Anyanwu et al., (2016).

The availability of firefighting equipment, training, and building occupants' awareness of its use were the main topics of a Tanzanian survey on fire preparedness. According to the study, a home's proximity to a fire department is vital for prompt emergency response. Márquez Sierra et al., (2012). Fire preparedness includes extinguishers, detectors, sprinklers, exit routes, and drills. It also involves building connections with local fire authorities, ensuring buildings are safe, and training people to use the equipment Hsiao & Hsieh, (2023); Jaafar et al., (2022). Together, these actions make the community better prepared and more resilient.

1.4 Schematic Diagram

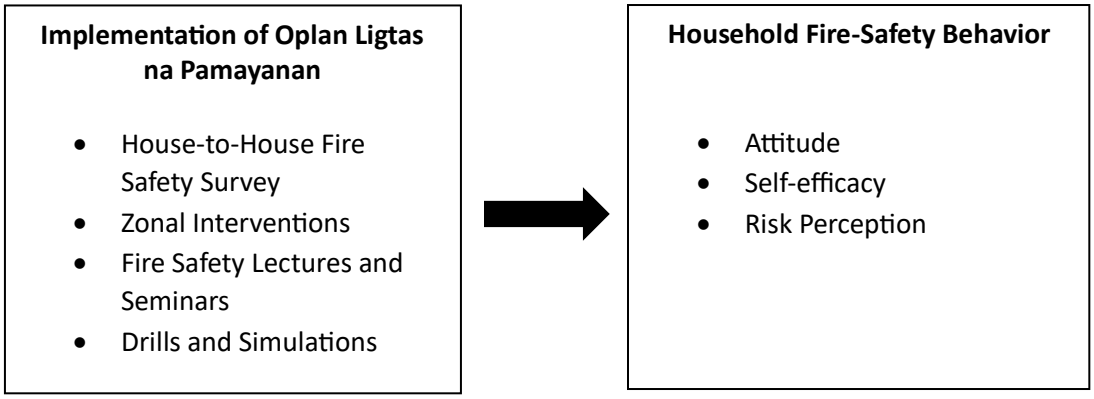


Figure 2. The Schematic Diagram shows the relationship between the dependent and independent variables of the study.

1.5 Statement of the Problem/ Objectives of the Study

This study aims to determine the impact of implementing **Oplan Ligtas na Pamayanan** on the **fire safety behavior of households**. Specifically, it seeks to answer the following questions:

1. **What is the level of implementation in Oplan Ligtas na Pamayanan** in the community in terms of:

    2.1 House-to-House Fire Safety Survey

    2.2 Zonal Interventions

    2.3 Fire Safety Lectures and Seminars

    2.4 Drills and Simulations

2. **What is the level of household fire-safety behavior** of the respondents in terms of:

    3.1 Attitude

    3.2 Self-efficacy

    3.3 Risk Perception

3. **Is there a significant relationship between implementation of Oplan Ligtas na Pamayanan and the household fire-safety behavior of the respondents?**

1.6 Hypothesis

The following hypothesis will guide the study.

H1: There is no significant relationship between the Profile of the Respondents and the implementation of the Oplan Ligtas na Pamayanan and the household fire-safety behavior.

1.7 Definition of Terms

Age. The number of years a person has lived is often used as a demographic factor in studies or surveys.

Attitude. The respondents' disposition, feelings, and tendencies toward fire safety were measured by how often they practice fire prevention measures and participate in preventive activities.

Bureau of Fire Protection (BFP). The government agency responsible for implementing Oplan Ligtas na Pamayanan (OLP) in Buenavista, Nasipit, and Carmen (BueNasCar). In this study, it refers specifically to the local BFP personnel who conducted inspections, drills, and educational campaigns, and whose actions are evaluated to determine program effectiveness.

Campaigns. Structured advocacy efforts designed to promote fire safety awareness and to encourage community members to participate in preventive activities, measured by attendance, engagement, and implementation of safety practices.

Community Fire Auxiliary Group (CFAG). A barangay-based volunteer group trained and supervised by the BFP to assist in fire prevention and response activities. In this study, CFAG members are considered local partners whose participation and activities are evaluated to support the implementation of Oplan Ligtas na Pamayanan (OLP).

Drills & Simulations. Practical exercises conducted in communities to train residents on how to respond appropriately during fire emergencies, measured by participation and adherence to safety procedures.

Fire Safety Behavior. The practices of households in Buenavista, Nasipit, and Carmen (BueNasCar) are influenced by the Oplan Ligtas na Pamayanan (OLP) program. This includes their awareness of fire hazards, safe use of electricity, participation in fire drills, and preparedness for emergencies,

such as having fire extinguishers and knowing evacuation routes. These behaviors are measured by observation and self-reported compliance with safety practices.

Fire Safety Education. The process of providing information, instruction, and awareness to individuals and households about fire risks, preventive measures, and appropriate responses during fire emergencies.

Fire Safety Knowledge. The level of awareness and understanding of the respondents regarding fire hazards, preventive measures, and appropriate responses during fire emergencies.

Fire Safety Lectures and Seminars. Refer to the educational sessions conducted by BFP officers aimed at increasing public awareness and knowledge about fire prevention, evacuation procedures, and safe household practices.

Gender. Individuals' social, cultural, and personal identification as male, female, or other identities. It is distinct from biological sex.

Households. Refers to the families or target respondents that participated in the Oplan Ligtas na Pamayanan program of the Bureau of Fire Protection in Buenavista, Nasipit, and Carmen, Agusan del Norte.

House-to-House Fire Safety Survey. Refers to the activity conducted by the Bureau of Fire Protection (BFP) personnel who visit each household to assess their compliance with basic fire safety standards.

Inspections. The systematic process of examining households, establishments, or community areas to identify potential fire hazards and ensure compliance with fire safety standards.

Oplan Ligtas na Pamayanan. The fire prevention program was designed and implemented by the Bureau of Fire Protection in Buenavista, Nasipit, and Carmen, Agusan del Norte.

Preparedness. In the context of this study, the readiness of the participating households to prevent and respond to fire incidents is examined. It is measured through their possession of fire safety equipment, established emergency plans, and participation in fire safety drills promoted by OLP.

Risk Perception. The respondents' awareness and personal judgment of the likelihood and potential severity of fire hazards in their homes and communities are being assessed.

Self-efficacy. The respondents' confidence in their ability to perform actions and apply safety measures that reduce fire risks and effectively respond during fire emergencies is crucial.

Zonal Intervention. Refers to the organized fire safety activities and initiatives implemented by the BFP within specific zones or clusters in a barangay.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter explains the methods used to achieve the study's goal. It also covers the research materials, the methods for analyzing the data collected, the research sample chosen from the population, and the methodology used as a guide for the study. The study also looked at ethical considerations and carefully reviewed the strengths and weaknesses of the methodology.

2.1 Research Design

This study uses a descriptive–quantitative research design, which, according to McCombes, (2023), systematically describes and measures the characteristics of a population or phenomenon without changing any variables. Using this method, the impact of the Oplan Ligtas na Pamayanan (OLP) program of the Bureau of Fire Protection (BFP) on households' fire safety practices will be evaluated. The primary data collection tool will be a structured questionnaire using a four-point Likert scale. The degree of respondents' knowledge, readiness, and preventive practices can be systematically measured through this design, which also enables statistical analysis to identify trends and connections.

Households directly exposed to OLP activities, such as community drills, fire safety lectures, and house-to-house inspections, will be purposively sampled. To investigate relationships between program participation and household fire-safety behavior, data will be analyzed using descriptive and inferential statistics, such as frequencies, percentages, weighted means, and, when applicable, Pearson Correlation tests.

2.2 Research Materials and Methods

A questionnaire developed by the researcher to determine households' fire safety practices under the Oplan Ligtas na Pamayanan (OLP) program will be the primary instrument for data collection. To capture three important dimensions—knowledge of fire-safety practices,

household preparedness, and preventive practices—the questionnaire will be designed with reference to the Bureau of Fire Protection's OLP Guidebook and pertinent academic literature on community-based fire prevention. BFP, (2022)

The first section will collect basic household and demographic information, such as age, household size, and the presence of fire safety equipment. To evaluate respondents' knowledge, frequency of preparedness activities, and consistency in implementing preventive measures, the following sections will include statements rated on a 4-point Likert scale, with response options ranging from 4 (Strongly Agree/Always) to 1 (Strongly Disagree/Never). The final questionnaire will include suggestions from academic reviewers and fire-safety experts from the Bureau of Fire Protection, who will check that each item correctly represents the intended concepts.

2.3 Research Environment, Population and Sample

The study will be conducted in the municipalities of Buenavista, Nasipit, and Carmen (BueNasCar) in Agusan del Norte. These nearby municipalities, which include both urban and rural barangays, have been designated as priority areas for community-based fire prevention initiatives because of their varied housing conditions and vulnerability to residential fire hazards.

Since 2019, the BFP has actively carried out the OLP program in these areas through community fire drills, educational campaigns, and house-to-house inspections. Each municipality has also formed a Community Fire Auxiliary Group (CFAG) to strengthen local emergency response and fire prevention. The respondents will be households from selected barangays in BueNasCar that have taken part in OLP activities.

A purposive sampling technique will be used to include only households that have directly participated in the program's fire safety activities. To make sure each question properly measures household knowledge, preparedness, and safety practices, the questionnaire will be reviewed by

experts from the SMCC College of Criminal Justice Education Department. To ensure clarity and applicability, revisions will be made based on expert input.

To further describe the research environment, the selected municipalities and barangays, along with their respective population and household data, are presented in Table 1.

Table 1. Respondents of the Study

Households	Population	Sample	
	F	f	%
Buenavista			
Barangay 7	253	19	5.3
Barangay 9	402	31	8.7
Barangay10	614	47	13.2
Nasipit			
Barangay 3	247	19	5.3
Barangay 7	67	5	1.4
Sta. Ana	636	48	13.5
Carmen			
Centro	1791	137	38.5
Cahayagan	658	50	14.0
Total	**4668**	**356**	**100.00**

2.4 Research Data and Analysis Plan

To determine the number of households in Buenavista, Nasipit, and Carmen, the researchers will first seek permission from the barangay officials. Following the acquisition of the respondent list, the researchers will meticulously organize the data collection procedure, ensuring that participants' rights, security, and benefits are protected.

The researchers will then give participants an informed consent form that thoroughly explains the goals, methods, and confidentiality precautions of the study. The consent form will

be available for community members to read over and sign, attesting to their comprehension and voluntary participation in the study.

After the researchers gather the informed consent forms, the questionnaire will be administered during in-person consultations to accommodate each respondent's preferences. After the questionnaires are distributed, the researchers will gather all of the completed questionnaires and use the selected statistical tool to analyze, total, verify, tabulate, and interpret the results.

After that, all the data will be organized and coded for statistical analysis. Following the analysis, the BFP and the relevant barangay councils will receive a summary of the results to support continued fire safety efforts.

The data will be analyzed using appropriate statistical methods:

Frequency and percentage distribution. To describe the demographic profile of respondents and their fire-safety practices.

Weighted mean. To determine the overall levels of knowledge, preparedness, and preventive practices based on the Likert-scale ratings.

Pearson Correlation. To examine the association between participation in OLP activities and household fire-safety behavior.

Table 1. Weighted Mean Scores Interpretation

Likert/Rating Scale	Mean Range/ Parameter		Description	Remarks/ Interpretation
	Lower Limit	Upper Limit		
4	3.25	4.00	Strongly Agree	High
3	2.50	3.24	Agree	Moderate
2	1.75	2.49	Disagree	Low
1	1.00	1.74	Strongly Disagree	Very Low

2.5 Research Ethical Standards

Prior to data collection, the researchers obtained written consent from the BFP and the relevant barangay councils and ethics clearance from the Institutional Research Ethics Committee. The goals, methods, and possible advantages of the study were explained to each participant. They were told that participation was entirely voluntary and that they could leave at any time without incurring any fees, and they signed a written informed consent form. The researchers kept all gathered information completely private, kept it in password-protected files, and, following the required retention period, disposed of it in compliance with institutional policies.

A. Intellectual Property Rights (IPR)

The research does not involve the development of new technologies, processes, or products that would necessitate patent protection. However, it includes original materials such as reports and surveys that require copyright protection to preserve the integrity of the work and prevent unauthorized reproduction or distribution. Additionally, the study does not involve the creation of any unique brand, logo, or symbol that would warrant trademark registration.

B. Informed Consent

Before the study began, participants were fully informed about its purpose, methods, possible risks, and benefits. The researchers explained the study's goals and procedures clearly to ensure participants fully understood them. All participants provided written consent via an informed consent form that guaranteed their rights as participants and the confidentiality of their answers. Since no animals or other living organisms were used in the study, ethical approval for animal or living-organism trials was not required.

C. Data Privacy and Confidentiality

Participants' personal information was collected and kept safe following the Philippines' Data Privacy Act of 2012 (RA 10173). To maintain confidentiality and privacy, all participant data was anonymized. Questionnaires and other data were kept in a locked drawer, and digital files were encrypted to block unauthorized access. Only the researchers had access to the data. Furthermore, unless participants specifically consent, anonymized data will not be shared with any third parties; any sharing will be for academic purposes only.

D. Voluntary Participation

Every human participant in the study was made fully aware of their right to discontinue participation at any moment without incurring any penalties, and participation was completely voluntary. To ensure ethical compliance and respect for participants' autonomy, the principles of voluntary participation were rigorously adhered to. Since there were no animal subjects or associated experimental procedures in the study, ethical guidelines for the use of animals were not relevant.

E. Minimization of Harm

A risk assessment was conducted to identify and reduce potential physical, psychological, or emotional harm to protect the welfare of human participants. Before beginning the study, participants received comprehensive information about their rights, including the freedom to discontinue participation at any time if they experienced discomfort or distress. Additionally, participants had access to support services if needed. The researchers did not use any animals in the study, so ethical considerations for animal participants did not apply.

F. Beneficence and Contribution to Knowledge

The study supports the beneficence principle by making a constructive contribution to society and the academic community. To enhance awareness campaigns and encourage

responsible safety practices, it examines how the Oplan Ligtas na Pamayanan program affects locals' fire safety behavior. The research's conclusions aim to promote a more proactive, fire-resilient community. Furthermore, participants will have access to the research findings upon request, guaranteeing openness and information exchange.

G. Justice and Fair Participant Selection

To ensure that all participants were included and excluded based on clear, equitable criteria, the study adhered to the principles of justice and fairness in participant selection. Equal opportunity to participate was ensured by including human participants from diverse backgrounds without discrimination. The researchers did not use any animals in this study, so ethical issues regarding animal testing did not apply.

H. Data Integrity and Accuracy

The research ensured data integrity and accuracy by employing standardized data collection and analysis procedures to ensure reliability and validity. All outputs were thoroughly checked to avoid errors and potential biases. Further, the research acknowledged its limitations, including potential biases and limited sample size, to maintain transparency and ensure the integrity of the research process.

I. Transparency and Honesty in Reporting

The study adheres to the principle of transparency in reporting by ensuring that all research findings will be presented truthfully and accurately. The researchers made every effort to maintain honesty in the interpretation and presentation of results. Furthermore, all copyrighted and patented materials used in the study were cited correctly and acknowledged to uphold academic integrity and give due credit to sources.

J. Use of Patented or Copyrighted Materials

Since no patented technologies were used in the research, the consent of patent holders was not required. To maintain academic integrity and comply with intellectual property laws, the study used copyrighted materials, including software and literature, and obtained the necessary permissions and acknowledgments.

K. Ethical Considerations for Animal and Human Trials

The researchers fully informed all participants of the potential risks and benefits, and the study complied with ethical standards for human research. Ethical clearance for animal use was not required, as no animals were involved in the study. The researchers safeguarded participants' rights and welfare throughout the research process by adhering to ethical standards.

L. Responsible Use of Artificial Intelligence and Related Technologies

To aid the research process, the study used ethical, responsible artificial intelligence tools. The study used Grammarly to ensure grammatical correctness and clarity. ChatGPT was used in the preliminary stage to help organize the researcher's thoughts. All AI-generated content was thoroughly examined and verified to prevent bias or misinterpretation and preserve data integrity. Additionally, throughout the study, the researchers used these technologies to maintain transparency and strengthen ethical standards.

M. Ethical Approval

The study received ethical approval from the Saint Michael College of Caraga Institutional Review Board. The ethics board approved after the researcher fulfilled all requirements and finished the evaluation process. This ensures that the study complies with accepted ethical guidelines for the use of human subjects in research.

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